EXCRETION

Pharmacokinetics of Drug Elimination

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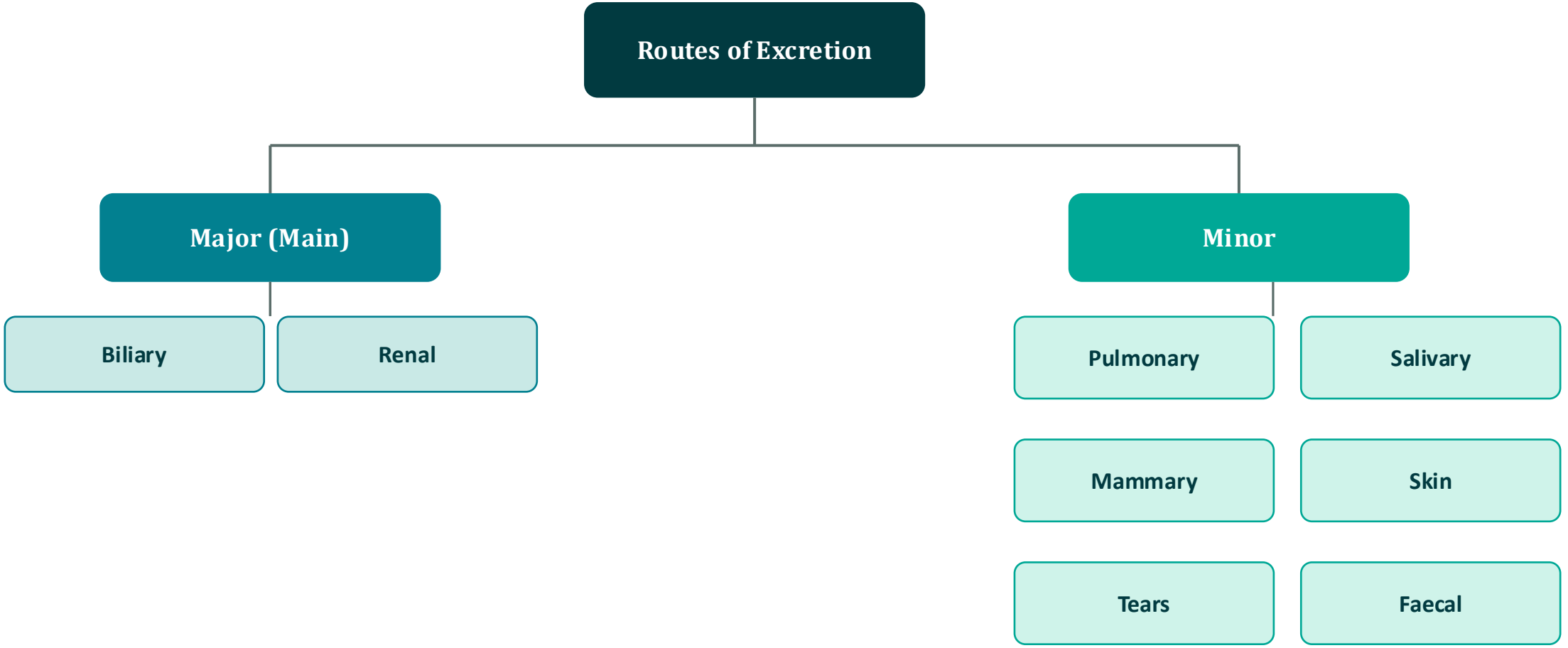
Objectives

1. Define drug excretion
2. List the different routes of excretion
3. Know the mechanism involved in each route of excretion

Definition

Excretion is defined as a process whereby drugs or metabolites are irreversibly transferred from the internal to the external environment through renal or non-renal routes.

Routes of Excretion



Drug Clearance by the Kidney

- Drugs must be sufficiently polar to be eliminated from the body. Removal of drugs from the body occurs via a number of routes; the most important is elimination through the kidney into the urine.
- Patients with renal dysfunction may be unable to excrete drugs and are at risk for drug accumulation and adverse effects.
- A drug passes through several processes in the kidney before elimination: glomerular filtration, active tubular secretion, and passive tubular reabsorption.

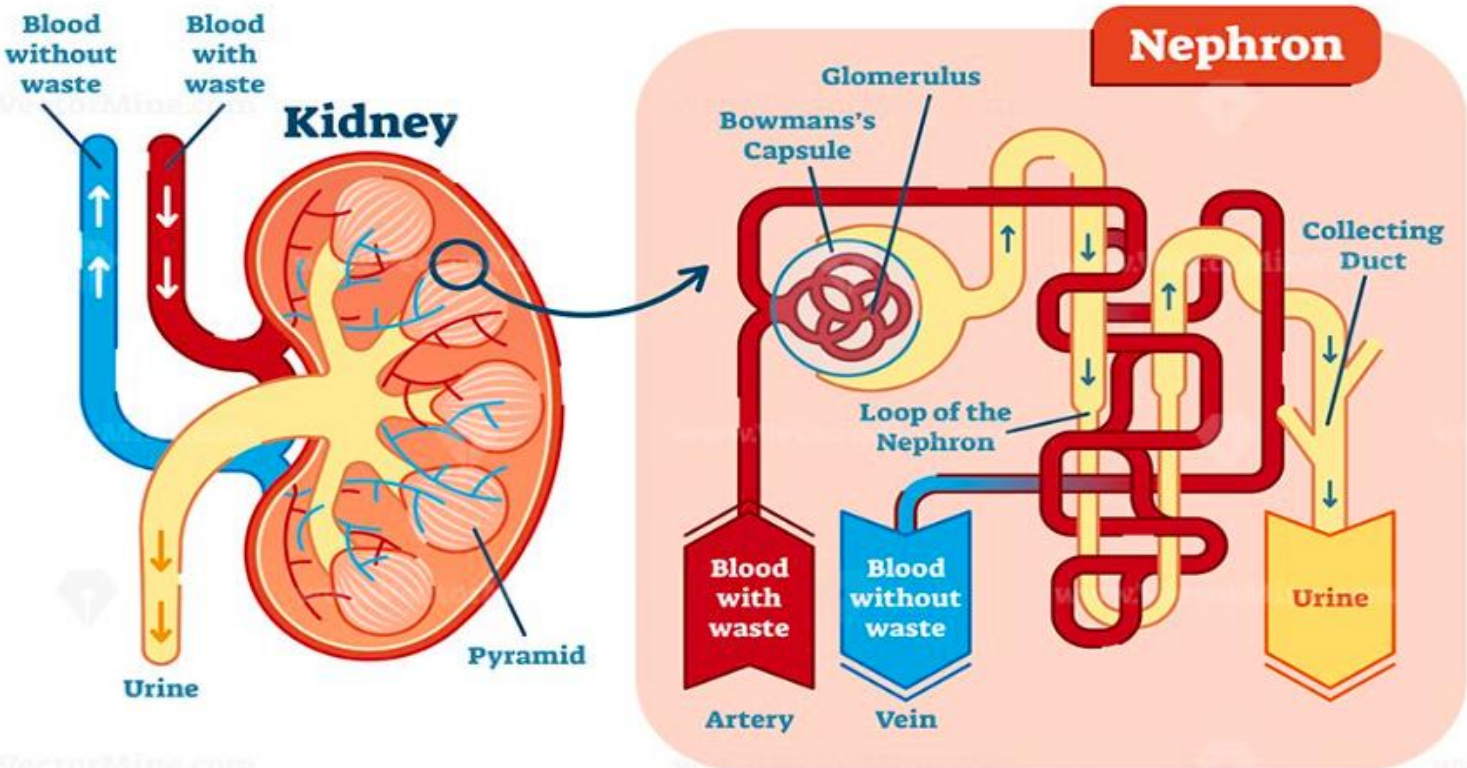
1. Glomerular Filtration

- Drugs enter the kidney through renal arteries, which divide to form a glomerular capillary plexus. Free drug (not bound to albumin) flows through the capillary slits into the Bowman space as part of the glomerular filtrate.
- The glomerular filtration rate (GFR) is normally about 120 mL/min/1.73m2 but may diminish significantly in renal disease.
- Lipid solubility and pH do not influence the passage of drugs into the glomerular filtrate. However, variations in GFR and protein binding of drugs do affect this process.

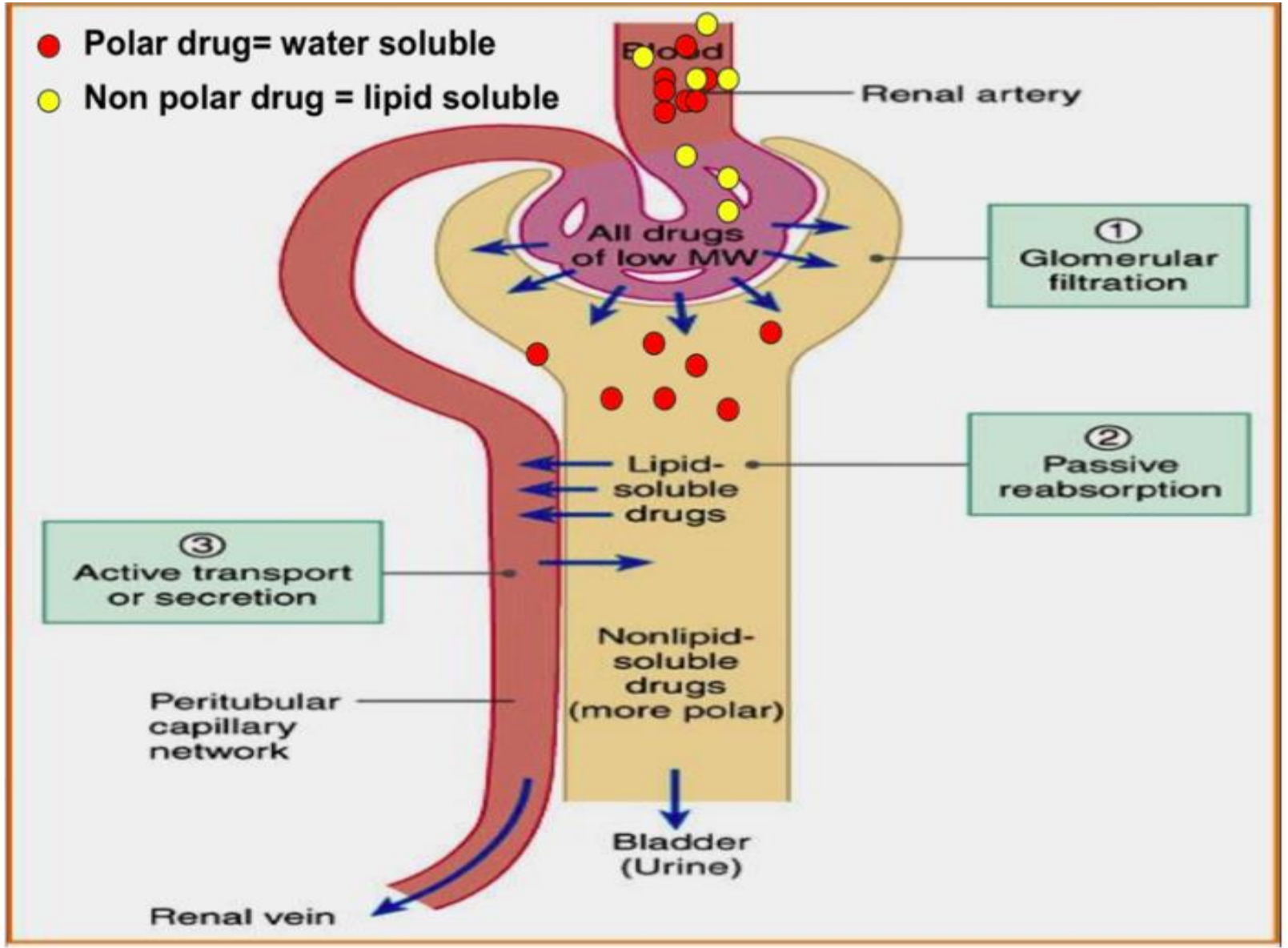
Glomerular Filtration — Molecular Weight & Binding

- Glomerular capillaries allow drug molecules of molecular weight below about 20 kDa to pass into the glomerular filtrate.
- Plasma albumin (MW about 68 kDa) is almost completely impermeant but most drugs cross the barrier freely.
- When a drug is bound to plasma proteins (warfarin 98%), only the free form is available for filtration.

Nephron Anatomy



Filtration, Reabsorption & Secretion



Key Point: Glomerular Filtration

In other words: glomerular filtration depends on plasma **protein binding** and **renal blood flow**. It does not depend on lipid solubility because all substances (whether water-soluble or lipid-soluble) can cross the fenestrated glomerular membrane.

2. Proximal Tubular Secretion

- Drugs that were not transferred into the glomerular filtrate leave the glomeruli through efferent arterioles, which divide to form a capillary plexus surrounding the nephric lumen in the proximal tubule.
- Secretion primarily occurs in the proximal tubules by two energy-requiring active transport systems: one for anions (also known as OAT) and one for cations (also known as OCT).

Transport Systems: OAT vs OCT

OAT (Organic Anion Transporter)

- p-Aminohippuric acid
- Furosemide
- Glucuronic acid conjugates
- Glycine conjugates
- Indometacin
- Methotrexate
- Penicillin
- Probenecid
- Sulfate conjugates
- Thiazide diuretics
- Uric acid

OCT (Organic Cation Transporter)

- Amiloride
- Dopamine
- Histamine
- Mepacrine
- Morphine
- Pethidine
- Quaternary ammonium compounds
- Quinine
- 5-Hydroxytryptamine (serotonin)
- Triamterene

Tubular Secretion — Efficiency

- Because at least 80% of the drug delivered to the kidney is presented to these carriers, tubular secretion is potentially the most effective mechanism of renal drug elimination.
- Unlike glomerular filtration, carrier mediated transport can achieve maximal drug clearance even when most of the drug is bound to plasma protein.

Tubular Secretion — Competition & Specificity

- Each of these transport systems shows low specificity and can transport many compounds. Thus, competition between drugs for these carriers can occur within each transport system.
- Note: Premature infants and neonates have an incompletely developed tubular secretory mechanism and, thus, may retain certain drugs in the blood.

Key Point: Tubular Secretion

In other words:
Tubular secretion does not depend on lipid solubility or plasma protein binding.

In the nephron, separate pumps are present for acidic and basic drugs.

Drugs utilizing the same transporters may show drug interactions e.g., probenecid decreases the renal excretion of **Penicillin** and increases the excretion of **uric acid**.

3. Distal Tubular Reabsorption

- As a drug moves toward the distal convoluted tubule, its concentration increases and exceeds that of the perivascular space. The drug, if uncharged, may diffuse out of the nephric lumen, back into the systemic circulation.
- Manipulating the urine pH to increase the fraction of ionized drug in the lumen may be done to minimize the amount of back diffusion and increase the clearance of an undesirable drug.

Ion Trapping

Generally, weak acids can be eliminated by alkalinization of the urine, whereas elimination of weak bases may be increased by acidification of the urine. This process is called ***"ion trapping."*** For example, a patient presenting with phenobarbital (weak acid) overdose can be given bicarbonate, which alkalinizes the urine and keeps the drug ionized, thereby decreasing its reabsorption.

Key Point: Tubular Reabsorption

- In other words: tubular reabsorption depends on lipid solubility. If a drug is lipid soluble, more of it will be reabsorbed and less will be excreted. Opposite is true for lipid insoluble drugs.
- As lipid solubility depends on ionization, the ionized drug will be excreted by the kidney.
- Thus in acidic drug poisoning (salicylates, barbiturates, chloramphenicol etc.) urine should be alkalinized with sodium bicarbonate.
- Similarly for basic drug poisoning (morphine, amphetamine etc.) urine should be acidified by using ammonium chloride.

Excretion by Other Routes

- Drug excretion may also occur via the intestines, bile, lungs, and breast milk, among others.
- Drugs that are not absorbed after oral administration or drugs that are secreted directly into the intestines or into bile are excreted in the faeces.

Faecal Elimination

- When any drug intended for systemic effect is taken by mouth a proportion may remain in the bowel and be excreted in the faeces.
- Some drugs are intended not be absorbed from the gut, so that it acts locally in the intestine e.g. Neomycin.

Faecal Elimination — Mechanisms

- The cells of the intestinal epithelium contain several carrier-mediated transporters that control the absorption of drugs.
- Drug in the blood may also diffuse passively into the gut lumen, depending on its pKa and the pH difference between blood and gut contents.
- The effectiveness of activated charcoal by mouth for drug over-dose depends partly on its adsorption of such diffused drug, and subsequent elimination in the faeces.

Biliary Excretion

- Transporters regulate the uptake of organic cations and anions from portal blood to hepatocyte, and thence to the bile.
- The bile canaliculi tend to reabsorb small molecules and in general, only compounds having a molecular weight greater than 300 pass into bile.

Pulmonary Excretion

- The lungs are the main route of elimination (and of up-take) of volatile anaesthetics.
- Apart from this, they play only a trivial role in drug elimination. The route, however, acquires notable medicolegal significance when ethanol concentration is measured in the air expired by vehicle drivers involved in road traffic accidents (via the breathalyser).

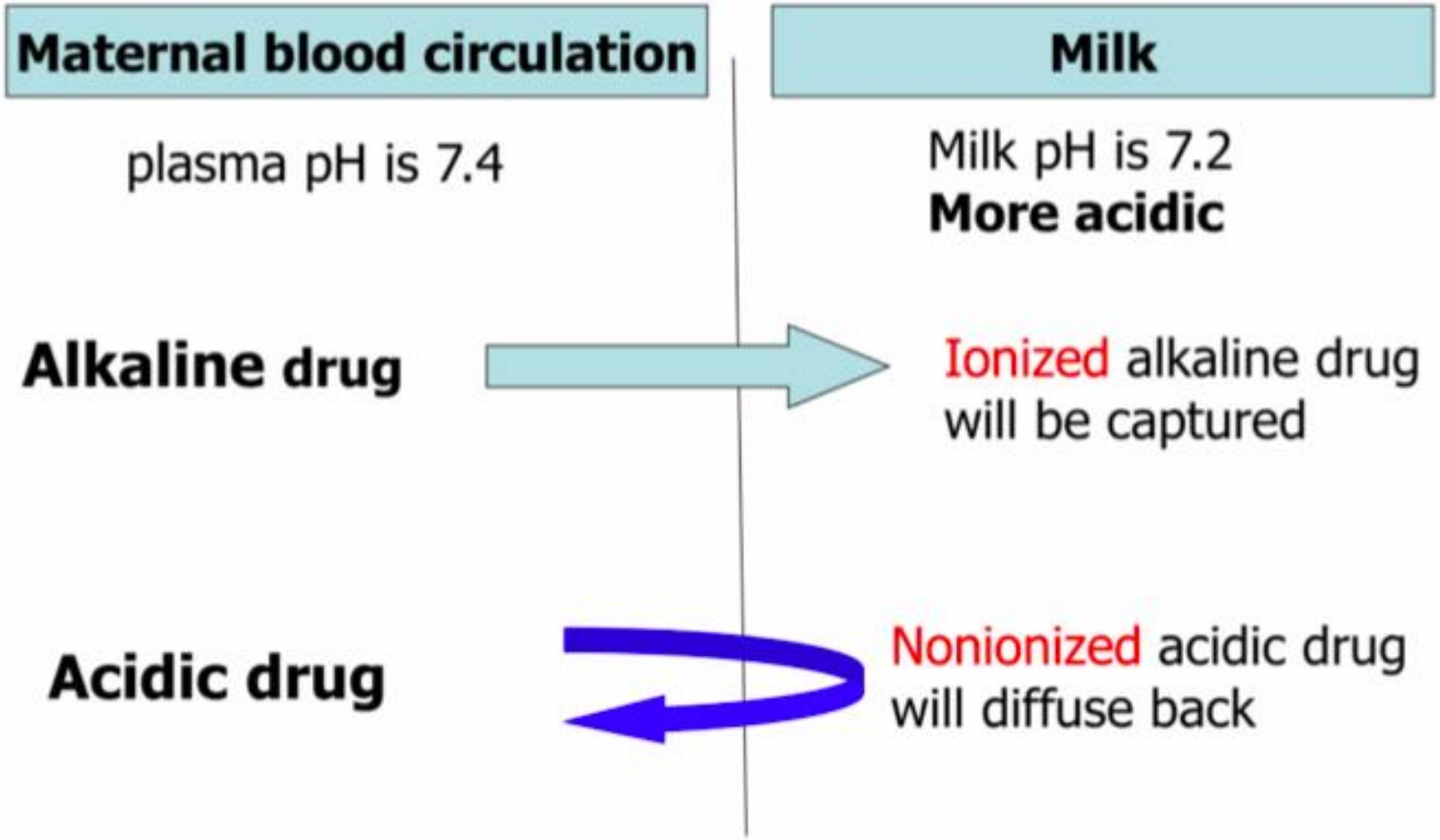


Breathalyser test

Mammary (Breast Milk) Excretion

- Most drugs that are present in a mother's plasma appear to some extent in her milk, although the amounts are so small that loss of drug in milk is of no significance as a mechanism of elimination.
- Even small amounts, however, may sometimes be of significance for the suckling child, whose drug metabolic and eliminating mechanisms are immature.
- While most drugs taken by the mother pose no hazard to the child, exceptions to this observation occur because some drugs are inherently toxic, or transfer to milk in significant amounts, or there are known adverse effects.

Effect of pH of the Plasma and Milk



Total Body Clearance

- The total body (systemic) clearance, CL total, is the sum of all clearances from the drug-metabolizing and drug-eliminating organs. The kidney is often the major organ of excretion. The liver also contributes to drug clearance through metabolism and/or excretion into the bile.

CL total = CL hepatic + CL renal + CL pulmonary + CL other

where CL hepatic + CL renal are typically the most important.

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